

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (CAD/CAM) Examination

May 2013

ME 710.01 Industrial Automation and Robotics

Date: 13.05.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q -1 (a) Compare Electrical, Hydraulic and Pneumatic systems [06]
(b) General Consideration of selection of Automation Methods [02]
- Q -2 (a) Explain Orientation of U-shape part and washer in vibratory-bowl feeder. [08]
(b) Explain continuous and intermittent transfer system [04]

OR

- Q -2 (a) What are the advantages of using Automated Guided Vehicle System (AGVS) in Flexible Manufacturing System (FMS). [06]
(b) What is ASRS? What are its types? [06]
- Q -3 Write short notes on any three: [15]
1. Balanced feeders
 2. Reciprocating-tube hopper feeder
 3. Gear Pump
 4. Lobed Pump
 5. Effect of Co-efficient of Friction and Frequency in vibratory conveyer

SECTION – II

- Q -4 (a) Explain Different specification of Robot in details. [05]
(b) Define: Resolution, Repatability and Accuracy. [03]
- Q -5 Write down following for a Stanford robotic manipulator [12]
- (i) Structure or configuration
 - (ii) Schematic diagram and joint co-ordinate system
 - (iii) Table of joint parameter
 - (iv) Overall direct kinematics transformation matrix

OR

Q-5 Write down following with respect to a standard six axis revolute robot:

[12]

1. Sketch structure,
2. Label joint co-ordinate system,
3. Prepare joint parameter table,
4. Write expression for direct kinematics.

Q-6 Write short notes on any three:

[15]

1. Geomtry Approach of Inverse Kinematics
2. Homogeneous transformation matrix for joint representation.
3. Linear Variable differential transformer (LVDT)
4. Rotary Encoder
5. Image processing for robotic vision systems.